Latihan 4.1

A

1. $f(t) = 2t - 1$

$F(s) = \frac{2}{s^2} - \frac{1}{s}$

2. $f(t) = e^{2t} - 1$

$F(s) = \frac{1}{s-2} - \frac{1}{s}$

B

7. $F(s) = \frac{1}{s} + \frac{1}{s^2}$

$\mathcal{L}^{-1}[F(s)] = 1 + t$

8. $F(s) = \frac{1}{s-2} - \frac{1}{s^3}$

$\mathcal{L}^{-1}[F(s)] = e^{2t} - \frac{t^2}{2}$

9. $F(s) = \frac{10}{s+3} - \frac{1}{s^2+4} = 10\left(\frac{1}{s+3}\right) - \frac{1}{s^2+4}$

$\mathcal{L}^{-1}(F(s)) = 10e^{-3t} - \frac{1}{2}\sin 2t$

Latihan 4.2

A

1. $f(t) = te^{2t}$

$F(s) = \frac{1}{(s-2)^2}$

2. $f(t) = e^{6t}\cos t$

$F(s) = \frac{s+6}{(s-6)^2+1}$

3. $f(t) = e^{4t}\sin 2t$

$F(s) = \frac{2}{(s-4)^2+4}$

B

6. $F(s) = \frac{3}{(s+2)^2} = 3\left(\frac{1}{(s+2)^2}\right)$

$\mathcal{L}^{-1}(F(s)) = 3te^{-2t}$

7. $F(s) = \frac{3}{(s-1)^2+4} = \frac{3}{2}\left(\frac{2}{(s-1)^2+4}\right)$

$\mathcal{L}^{-1}(F(s)) = \frac{3}{2}e^{t}\sin 2t$